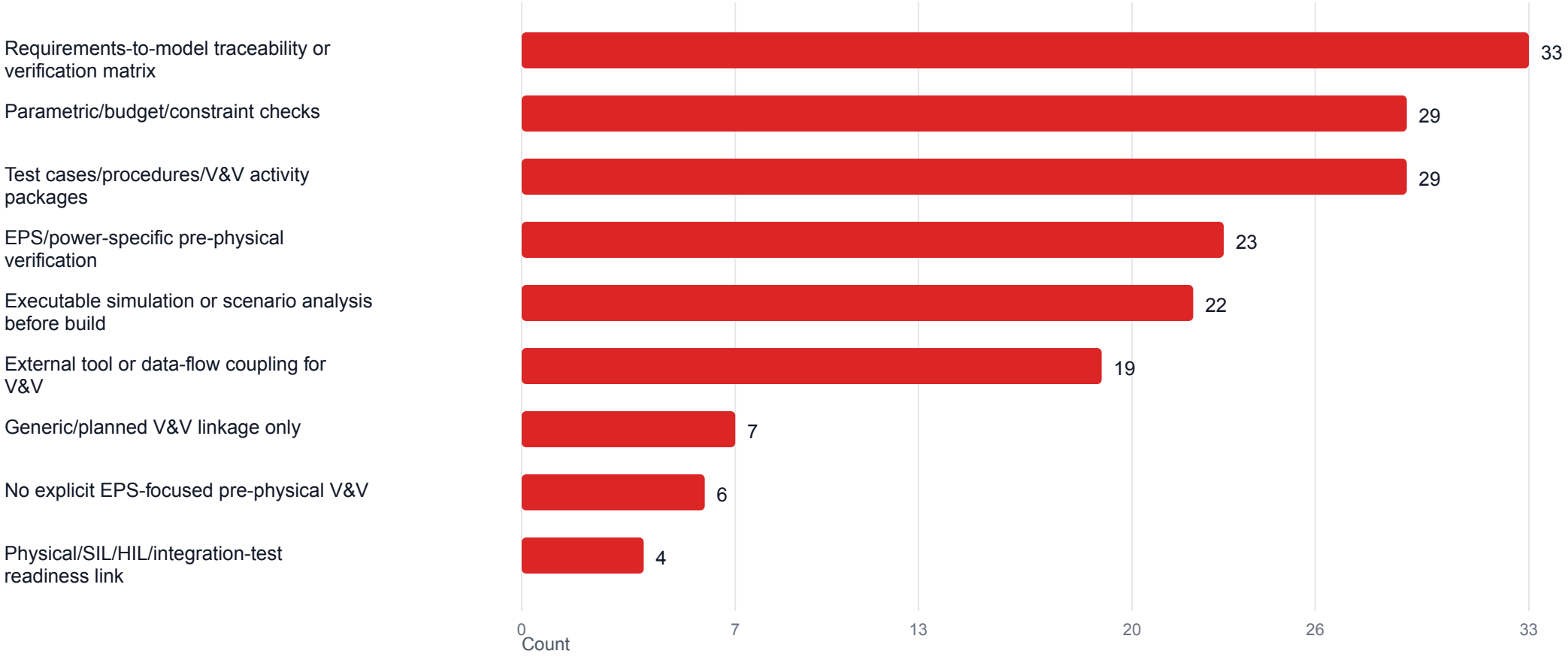Pre-physical V&V linkage categories